
Financial Management

Section - A

Cost of Capital

What is Cost of Capital (COC)?

Cost of Capital (COC):

Cost of Capital (COC) is the minimum required rate of return that company must earn to keep its stock price unchanged.

Overall Cost of Capital of the Firm :

- Cost of Capital is the required rate of return on the various types of financing.
- The overall cost of capital is a weighted average of the individual required rates of return (costs).

Significance of Cost of Capital

- Creation & Measurement of Value
- Financing Decisions – Optimal Capital Structure
- Capital Investment Decisions
- Leasing vs. borrowing to buy

Cost of Debt (Debentures)

Cost of Debt is the required rate of return on investment of the lenders of a company.

$$V_B = \sum_{t=1}^T \frac{C_t}{(1 + kd)^t} + \frac{\text{Face Value}}{(1 + kd)^T}$$

$$kd = \frac{C(1-T) + \frac{(F-P)}{n}}{0.6P + 0.4F}$$

Where

C = Coupon

F = Face Value

P = Price

n = years to maturity

Cost of Debt (Debentures)

Crux Limited has recently made an issue of debentures for Rs.400 lakh. Each debenture has a face value of Rs.100 and carries a rate of interest of 14 percent. The interest is payable annually and the maturity of the 10 years. If Crux Limited realizes Rs.97 per debenture and the corporate tax rate is 50 percent, what is the cost of the debenture to the company?

Cost of Debt (Debentures)

Issue of Debenture: at Par, Premium and Discount

The debentures may be issued:

- A. At Par
- B. At a Discount
- C. At a Premium

A. Issue of Debentures at Par:

Debentures are said to be issued at par when the amount collected for it is equal to the nominal value (face value) of the debentures; for example, issue of Rs. 1,000 debenture for Rs. 1,000.

B. Issue of Debentures at Discount:

When debentures are issued by company at a price less than its nominal value (face value) it is said to be issued at discount. It is important to note that the Companies Act has not put any restriction on the maximum limit of discount. For example, if a debenture of Rs. 1,000 is offered to public at Rs. 950, it is issue at a discount. Here Rs. 50 on each debenture is loss to the company.

Cost of Debt (Debentures)

Issue of Debenture: at Par, Premium and

The debentures may be issued:

- A. At Par
- B. At a Discount
- C. At a Premium

$$k_d = \frac{C(1-T) + \frac{(F-P)}{n}}{0.6P + 0.4F}$$

C. Issue of Debentures at Premium:

If debentures are issued at a price more than its nominal value (face value) such an issue is called issue at a premium. For example, if a debenture of Rs. 1000 is offered at 1,050, it is a case of issue of debentures at premium. The excess of issue price over face value is premium.

Cost of Debt (Debentures)

A company issues Rs. 1,00,000, 15% Debentures of Rs. 100 each. The company is in 40% tax bracket. You are required to compute the cost of debt after tax, if debentures are issued at (i) Par, (ii) 10% discount, and (iii) 10% premium

Cost of Debt (Debentures)

A company issues Rs. 1,00,000, 15% Debentures of Rs. 100 each. The company is in 40% tax bracket. You are required to compute the cost of debt after tax, if debentures are issued at (i) Par, (ii) 10% discount, and (iii) 10% premium

Cost of Term Loan

The cost of the term loans will be simply equal to the interest rate multiplied by (1 – tax rate).

The interest rate to be used here will be the interest rate applicable to the new term loan.

The interest is multiplied by (1 – tax rate) as interest on term loans is also tax deductible.

$$k_t = I (1 - T)$$

Where,

I = Interest rate

T = Tax rate.

Cost of Preferred Stock

The cost of a redeemable preference share (k_p) is defined as that discount rate which equates the proceeds from preference capital issue to the payments associated with the same i.e. dividend payment and principal payments, which can be

$$P = \sum_{t=1}^n \frac{P_D}{(1+k_p)^t} + \frac{F}{(1+k_p)^n}$$

where,

k_p = cost of preference capital

P_D = preference dividend per share payable annually

F = redemption price

P = net amount realized per share and n = maturity period

Cost of Preferred Stock

Cost of Debt is the required rate of return on investment of the lenders of a company.

$$k_p = \frac{P_D + \frac{(F-P)}{n}}{0.6P + 0.4F}$$

where,

k_p = cost of preference capital

P_D = preference dividend per share payable annually

F = redemption price

P = net amount realized per share and

n = maturity period

Cost of Preferred Stock

The preference share of Jubilant Ltd has a face value of Rs.100 and carries a dividend rate of 14 percent payable annually. The share is redeemable after 12 years at par. If the net amount realized per share is Rs.95, what is the cost of the preference capital?

Cost of Preferred Stock

Omega Enterprises issued 10 year, 9 percent preference shares four years ago. The preference share which has a face value of ₹ 100 is currently selling for ₹ 92. What is the cost of preference shares?

Cost of Equity Approaches (External Equity)

- Dividend Discount Model
- Capital-Asset Pricing Model
- Earnings-Price Ratio Approach

Cost of Equity Approaches (External Equity)

- Dividend Discount Model

The cost of equity capital, k_e , is the discount rate that equates the present value of all expected future dividends with the current market price of the stock.

$$P_0 = \frac{D_1}{(1+k_e)^1} + \frac{D_2}{(1+k_e)^2} + \dots + \frac{D_\infty}{(1+k_e)^\infty}$$

The constant dividend growth assumption reduces the model to:

$$k_e = (D_1 / P_0) + g$$

Assumes that dividends will grow at the constant rate “g” forever.

Determination of the Cost of Equity Capital (External Equity)

Assume that RIL has common stock outstanding with a current market value of Rs.64.80 per share, current dividend of Rs.3 per share, and a dividend growth rate of 8% forever.

Determination of the Cost of Equity Capital (External Equity)

$$k_e = .05 + .08 = .13 \text{ or } 13\%$$

Capital Asset Pricing Model (External Equity)

The cost of equity capital, k_e , is equated to the required rate of return in market equilibrium. The risk-return relationship is described by the Security Market Line (SML).

$$k_e = R_j = R_f + (R_m - R_f) \beta$$

Whereas

$k_e = R_j$ = Cost of equity

R_f = Risk free Rate of Return

R_m = Market Return

β = Beta of the stock

The required rate of return is the minimum rate of return necessary to induce investors to buy or hold a security

Determination of the Cost of Equity (CAPM) (External Equity)

Assume that RIL has a company beta of 1.25. Research by Julie Miller suggests that the risk-free rate is 4% and the expected return on the market is 11.2%

Determination of the Cost of Equity (CAPM) (External Equity)

$$k_e = 4\% + 9\% = 13\%$$

Earnings-Price Ratio Approach (External Equity)

- Earnings-Price Ratio Approach

According to this approach, the cost of equity is equal to:

$$k_e = (E_1 / P_0)$$

where

E_1 = Expected earnings per share for the next year and

P_0 = Current market price per share.

E_1 may be estimated as: (Current earnings per share) $\times$ (1 + growth rate of earnings per share)

Cost of Equity Approaches (Internal Equity) : Retained Earnings

- Cost of Retained Earnings:
- The cost of retained earnings refers to the cost associated with a company financing new projects or investments using its retained earnings, rather than raising new equity or taking on debt. Retained earnings are the portion of a company's profits that are not distributed as dividends but are kept by the company to reinvest in its core business or to pay off its debts.
- The cost of retained earnings is essentially an opportunity cost, representing the return shareholders could have earned if they had received the retained earnings as dividends and invested them elsewhere.
- It's also seen as the return shareholders expect for the reinvestment of earnings back into the company.

Cost of Equity Approaches (Internal Equity) : Retained Earnings

- When a company retains its earnings instead of distributing them as dividends, there is an implicit cost associated with shareholders missing out on potential returns from those funds.
- In case the funds are retained, the shareholders won't receive those funds as dividend.
- So, this is the implicit cost that refers to the value that shareholders are foregoing by not receiving dividends.

Flotation Costs

- When a firm raises finance by issuing equity and debt, it almost invariably incurs floatation or issue costs, comprising of items like underwriting costs, brokerage expenses, fees of merchant bankers, advertising expenses, under-pricing cost, so on and so forth.
- For example, the floatation costs associated with equity issues may be 6 to 10 percent.

Concept of weighted average cost of capital

Dabur Ltd. has the following capital structure: (Rs.in lakhs)

Equity Capital (10 lakh shares at par value)	100
12 percent preference capital (10,000 shares at par value)	10
14% Non-convertible Debentures (70,000 debentures at par value)	70
14% term loan from APSFC	100
Total	280

The market price per equity share is Rs.25. The next expected dividend per share (DPS) is Rs.2.00 and the DPS is expected to grow at a constant rate of 8 percent. The preference shares are redeemable after 7 years at par and are currently quoted at Rs.75 per share on the stock exchange. The debentures are redeemable after 6 years at par and their current market quotation is Rs.90 per share. The tax rate applicable to the firm is 50 percent. Calculate the weighted average cost of capital.

Concept of weighted average cost of capital

The capital structure of Adamus Ltd. in book value terms is as follows: (Rs in millions)

Equity capital (20 million shares, ₹ 10 par)	200
Preference capital, 12 percent (500,000 shares, ₹ 100 par)	50
Debentures 14 percent (1,200,000 debentures, ₹ 100 par)	120
Term loans, 13 percent	80
Total Capital	450

The next expected dividend per share is ₹ 2.00. The dividend per share is expected to grow at the rate of 12 percent. The market price per share is ₹ 50.00. Preference stock, redeemable after 10 years, is currently selling for ₹ 85.00 per share. Debentures, redeemable after 5 years, are selling for ₹ 90.00 per debenture. The tax rate for the company is 30 percent. Calculate the average cost of capital.

WACC for Book Value and Market Value

The following is the Capital Structure of ABC Limited

Sources of Capital	Book Value	Market Value	Cost of Capital (After Tax)
Debentures	50,00,000	50,00,000	8.5
Preference Share Capital	40,00,000	40,00,000	11.25
Equity Share Capital	60,00,000	10500000	16.5

Calculate the Weighted Average cost of capital using A) Book Value and B) Market Values

Thank You